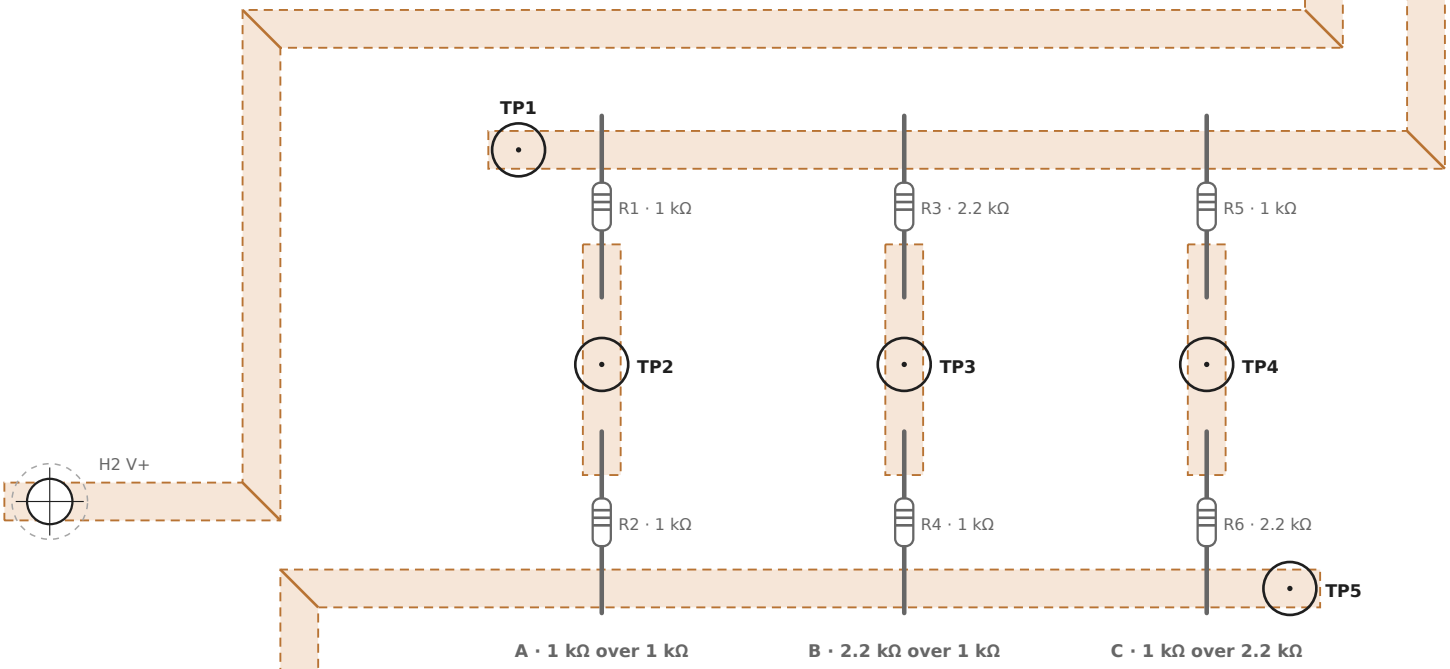
L07 Share the voltage

Two resistors in a row split the supply between them. Find the rule.

Draft: tagged steps may change

H1 BUS A · NC

S1
fold the corner down



H2 V+

H3 GND

You need

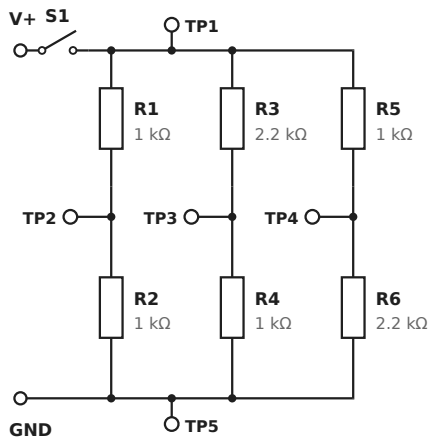
- 4 resistors, 1 kΩ (brown-black-red)
- 2 resistors, 2.2 kΩ (red-red-red)
- 5 mm copper tape
- Power page P1 in the binder
- A multimeter

Build it

1. Lay the six lanes, one piece of tape each. Fold at the creases; never cut.
2. Stick a 10 mm piece of copper tape on the corner flap, S1. **[T0]**
3. Punch the holes and set eyelets in H2 and H3. **[T0]**
4. Sort the resistors by colour: four 1 kΩ and two 2.2 kΩ. Place each one on its labelled gap, either way round.
5. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
6. Bind this page with P1. Hold the corner down while you measure.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Schematic



A is R1 and R2, B is R3 and R4, C is R5 and R6.

Predict Which middle point will read the highest voltage?

Highest A / B / C

Circle one. Which resistor do you think gets the bigger share, and why?

Measure Hold the corner down. Black probe on TP5.

Middle	Measured	Worked out
A TP2	____ V	____ V
B TP3	____ V	____ V
C TP4	____ V	____ V

TP1 ____ V

Worked out =
TP1 × bottom
÷ (top + bottom)

Explain What happens to the middle when you swap top and bottom?

Compare B and C. Which pair would give you about 3 V from P1?

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.